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Midterm 1 (Max. score = 80 points)

Math 589B, Spring 2024

Q 1. (10 points) Given is the following Newton's divided difference table of the function $f(x)$:

x_i	$f[x_i]$	$f[x_i, x_{i+1}]$	$f[x_i, x_{i+1}, x_{i+2}]$	$f[x_i, x_{i+1}, x_{i+2}, x_{i+3}]$	$f[x_i, x_{i+1}, x_{i+2}, x_{i+3}, x_{i+4}]$
0	0	0.7956	0.1312	0.2379	0.1792
0.25	0.1989	0.8612	0.3096	0.4171	
0.50	0.4142	1.016	0.6224		
0.75	0.6682	1.3272			
1.00	1				

Use the 4th-degree Newton polynomial constructed from the table to approximate $f(0.1)$. Make sure to give the answer with the correct number of significant digits, assuming that the values in the table are correctly rounded to 4 significant digits, i.e., have relative error $\leq 0.5 \times 10^{-4}$ (1/2 ulp guarantee).

Solution:

$$\begin{aligned}
 f(0.1) &\approx \\
 &f[0] \\
 &+ f[0, 0.25] (0.1 - 0) \\
 &+ f[0, 0.25, 0.50] (0.1 - 0) (0.1 - 0.25) \\
 &+ f[0, 0.25, 0.50, 0.75] (0.1 - 0) (0.1 - 0.25) (0.1 - 0.50) \\
 &+ f[0, 0.25, 0.50, 0.75, 1] (0.1 - 0) (0.1 - 0.25) (0.1 - 0.50) (0.1 - 0.75) \\
 &= 0 + \underbrace{0.7956 (0.1)}_{0.07956} + \underbrace{0.1312 (0.1)(-0.15)}_{0.01312} + \underbrace{0.2379 (0.1)(-0.15)(-0.40)}_{0.0014274} \\
 &+ \underbrace{0.1792 (0.1)(-0.15)(-0.40)(-0.65)}_{-0.00069888} \\
 &= 0.07832052 \approx 0.07832.
 \end{aligned}$$

The result of the computation is exact, but every divided difference has 4 digits after the decimal point, and the 1/2-ulp guarantee assumes that all divided difference approximations are within $0.5 \cdot 10^{-4}$ of the exact values. Since the first value in the table is multiplied by 0.1, which is known exactly. Therefore $0.7956 \cdot 0.1 = 0.07956$ has 6 accurate digits, i.e., 5 accurate digits after the decimal point.

The other quantities in the sum have negligible impact on accuracy of the sum because

they are by at least an order of magnitude smaller. Hence, we should round the answer to 5 digits after the decimal point.

THE ANSWER:

0.07832, rounded to **five digits** after the decimal point

- Q 2.** In numerical integration, the Midpoint Rule (for a single subinterval) approximates the definite integral of a function f over $[a, b]$ by evaluating f at the midpoint $m = \frac{a+b}{2}$:

$$\int_a^b f(x) dx \approx (b-a) f\left(\frac{a+b}{2}\right).$$

- Q 2.1.** (10 points) Write down (in your own words) the statement of the Midpoint Rule for approximating an integral of a continuous function f over the interval $[a, b]$.

Proof: The formula results from approximating the area by a rectangle of height $f(m)$ and base $(b-a)$. Drawing in the book.

- Q 2.2.** (10 points) Using ideas from how the error was derived for the Trapezoidal Rule (which we discussed in class), derive an error expression of the form

$$\text{Error} = -\frac{(b-a)^3}{24} f''(\xi)$$

for some $\xi \in (a, b)$. (Hint: consider the second derivative of f , apply an appropriate form of the Remainder Theorem or Taylor expansion, and carefully compare the exact integral to the midpoint approximation.)

Solution: We use Taylor expansion about the middle point m :

$$f(x) = f(m) + f'(m)(x-m) + \frac{f''(\xi_x)}{2}(x-m)^2.$$

Integrating over x over $[a, b]$ gives

$$\begin{aligned} \int_a^b f(x) dx &= f(m)(b-a) + f'(m) \overbrace{\int_a^b (x-m) dx}^0 + \int_a^b \frac{f''(\xi_x)}{2} (x-m)^2 dx \\ &= f(m)(b-a) + \frac{f''(\xi)}{2} \int_a^b (x-m)^2 dx. \end{aligned}$$

In the last step we applied the Mean Value Theorem for integrals. Finally,

$$\begin{aligned}
 \int_a^b (x - m)^2 dx &= \frac{1}{3}(x - m)^3 \Big|_{x=a}^b \\
 &= \frac{1}{3}((b - m)^3 - (a - m)^3) \\
 &= \frac{1}{3} \left(\left(\frac{b - a}{2} \right)^3 - \left(\frac{a - b}{2} \right)^3 \right) \\
 &= \frac{1}{3} \left(\left(\frac{b - a}{2} \right)^3 + \left(\frac{b - a}{2} \right)^3 \right) \\
 &= \frac{1}{3} \left(\left(\frac{b - a}{2} \right)^3 + \left(\frac{b - a}{2} \right)^3 \right) \\
 &= \frac{2}{3} \left(\frac{b - a}{2} \right)^3 \\
 &= \frac{1}{12}(b - a)^3.
 \end{aligned}$$

Combining with the previous, we obtain:

THE ANSWER:

$$\text{Error} = \frac{f''(\xi)}{24}(b - a)^3$$

NOTE: In particular, when f is convex, the error is positive, which means that the Mid-point Rule underestimates the integral (the opposite of the Trapezoid Rule).

- Q 2.3.** (10 points) Give a detailed proof that shows why there must exist such a point ξ in (a, b) satisfying the error bound. (You may use the Mean Value Theorem for integrals, Rolle's Theorem or other standard theorems covered in class and/or calculus.)

Proof: The standard “proof” is given in the derivation of the previous part. However, the proof is known to have a gap, because ξ_x may not depend continuously on x .

- Q 3.** Consider the Legendre polynomial of degree 3, $P_3(x)$. By definition, Legendre polynomials can be obtained from the *Rodrigues formula*:

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} [(x^2 - 1)^n].$$

- Q 3.1.** (10 points) Use the Rodrigues formula with $n = 3$ to derive an explicit expression for $P_3(x)$.

Solution: We start with the Rodrigues formula:

$$P_3(x) = \frac{1}{2^3 3!} \frac{d^3}{dx^3} [(x^2 - 1)^3] = \frac{1}{8 \cdot 6} \frac{d^3}{dx^3} [(x^2 - 1)^3].$$

First, expand $(x^2 - 1)^3 = (x^2 - 1)^2 (x^2 - 1) = (x^4 - 2x^2 + 1)(x^2 - 1)$. One may also expand directly:

$$(x^2 - 1)^3 = x^6 - 3x^4 + 3x^2 - 1.$$

Then,

$$\frac{d^3}{dx^3} (x^6 - 3x^4 + 3x^2 - 1) = \frac{d^3}{dx^3} (x^6) - 3 \frac{d^3}{dx^3} (x^4) + 3 \frac{d^3}{dx^3} (x^2) - \frac{d^3}{dx^3} (1).$$

Compute term by term:

$$\frac{d^3}{dx^3} (x^6) = 6 \cdot 5 \cdot 4 x^3 = 120 x^3, \quad \frac{d^3}{dx^3} (x^4) = 4 \cdot 3 \cdot 2 x = 24 x,$$

$$\frac{d^3}{dx^3} (x^2) = 2 \cdot 1 \cdot 0 x^0 = 0, \quad \frac{d^3}{dx^3} (1) = 0.$$

Hence,

$$\frac{d^3}{dx^3} [(x^2 - 1)^3] = 120 x^3 - 3(24 x) = 120 x^3 - 72 x = 24(5x^3 - 3x).$$

Therefore,

$$P_3(x) = \frac{1}{48} [24(5x^3 - 3x)] = \frac{5x^3 - 3x}{2}.$$

THE ANSWER:

$$P_3(x) = \frac{5x^3 - 3x}{2}$$

- Q 3.2.** (10 points) Find the roots of $P_3(x)$. These roots will be the *nodes* $x_1 < x_2 < x_3$ for the 3-point Gauss–Legendre quadrature rule on the interval $[-1, 1]$.

Solution: Set $P_3(x) = 0$. Then

$$\frac{5x^3 - 3x}{2} = 0 \implies x(5x^2 - 3) = 0.$$

This gives the solutions

$$x = 0, \quad x = \pm \sqrt{\frac{3}{5}}.$$

Thus, the three nodes are

$$x_1 = -\sqrt{\frac{3}{5}}, \quad x_2 = 0, \quad x_3 = \sqrt{\frac{3}{5}}.$$

Q 3.3. (10 points) The standard formula for the *weights* w_i in Gauss–Legendre quadrature is

$$w_i = \frac{2}{(1 - x_i^2) [P'_n(x_i)]^2},$$

where $P'_n(x)$ is the derivative of the Legendre polynomial of degree n . Compute the weights w_1, w_2, w_3 for $n = 3$, using the roots x_i from part (b).

Solution: We use the formula

$$w_i = \frac{2}{(1 - x_i^2) [P'_3(x_i)]^2}.$$

First, differentiate $P_3(x)$:

$$P'_3(x) = \frac{d}{dx} \left(\frac{5x^3 - 3x}{2} \right) = \frac{15x^2 - 3}{2} = \frac{3(5x^2 - 1)}{2}.$$

Weight at $x_2 = 0$.

$$P'_3(0) = \frac{3(5 \cdot 0^2 - 1)}{2} = -\frac{3}{2}.$$

Hence,

$$w_2 = \frac{2}{(1 - 0^2) \left(-\frac{3}{2}\right)^2} = \frac{2}{1 \cdot \frac{9}{4}} = \frac{2}{\frac{9}{4}} = \frac{8}{9}.$$

Weights at $x_1 = -\sqrt{\frac{3}{5}}$ and $x_3 = \sqrt{\frac{3}{5}}$.

By symmetry, these weights are equal. Let $x_+ = \sqrt{\frac{3}{5}}$. Then

$$1 - x_+^2 = 1 - \frac{3}{5} = \frac{2}{5}, \quad P'_3(x_+) = \frac{3}{2} \left(5 \cdot \frac{3}{5} - 1 \right) = \frac{3}{2} (3 - 1) = \frac{3}{2} \cdot 2 = 3.$$

Thus

$$[P'_3(x_+)]^2 = 9, \quad w_1 = w_3 = \frac{2}{\left(\frac{2}{5}\right) 9} = \frac{2}{\frac{18}{5}} = \frac{2 \cdot 5}{18} = \frac{5}{9}.$$

Q 3.4. (10 points) List your final answers for the *nodes* and *weights* $\{(x_1, w_1), (x_2, w_2), (x_3, w_3)\}$ in ascending order of the x_i . Hence, your completed 3-point Gauss–Legendre rule will have the form

$$\int_{-1}^1 f(x) dx \approx \sum_{i=1}^3 w_i f(x_i).$$

Solution: THE ANSWER:

$$\begin{aligned} x_1 &= -\sqrt{\frac{3}{5}}, & w_1 &= \frac{5}{9}, \\ x_2 &= 0, & w_2 &= \frac{8}{9}, \\ x_3 &= \sqrt{\frac{3}{5}}, & w_3 &= \frac{5}{9}. \end{aligned}$$